

Getting Started: VSCodium, Python, and GitHub/GitLab

M2 EMN Neurosciences 2026–2027 — Introduction to Programming

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This tutorial is written for complete beginners. Follow the steps in order — by the end you will have a working code editor, a working Python installation, and an account on GitHub or GitLab that you can use to store and share your code.

Part 1 — Installing VSCodium and Python

1.1 What are these tools?

- **VSCodium** is a free, open-source code editor (a version of Visual Studio Code without Microsoft's telemetry/branding). It is where you will write and run your Python scripts.
- **Python** is the programming language we will use in this course. It needs to be installed separately from the editor — VSCodium is just the tool you use to write the code, Python is what actually runs it.

1.2 Install Python

Windows

1. Go to python.org/downloads and download the latest Python 3 installer.
2. Run the installer. **Important:** on the first screen, tick the box "**Add python.exe to PATH**" at the bottom before clicking "Install Now". This step is easy to miss and causes most beginner problems later.
3. Once installed, open a terminal (search for "PowerShell" or "cmd" in the Start menu) and type:

```
python --version
```

You should see something like **Python 3.12.x**. If you get an error, restart your computer and try again.

macOS

1. Go to python.org/downloads and download the macOS installer, or install via [Homebrew](#) with `brew install python`.
2. Open the Terminal app and check the installation:

```
python3 --version
```

Linux

Python 3 is usually pre-installed. Check with:

```
python3 --version
```

If it's missing, install it with your package manager, e.g. `sudo apt install python3` (Debian/Ubuntu).

1.3 Install VSCode

1. Go to the official VSCode releases page: github.com/VSCodium/vscodium/releases
2. Download the installer for your operating system:
 - Windows: the `.exe` file (e.g. `VSCodiumUserSetup-x64-*.exe`)
 - macOS: the `.dmg` file (choose `arm64` if you have an Apple Silicon Mac, `x64` for an Intel Mac)
 - Linux: the `.deb/.rpm/.AppImage` matching your distribution
3. Run the installer and follow the on-screen instructions (default options are fine).
4. Launch VSCode once installed.

1.4 Set up VSCode for Python

1. Open VSCode.
2. Click the **Extensions** icon in the left sidebar (looks like four squares).
3. Search for **"Python"** (published by Microsoft) and click **Install**.
4. Also install the **"Jupyter"** extension the same way — we will use Jupyter notebooks (`.ipynb` files) during the course.
5. To test everything works:
 - Create a new folder on your computer for this course (e.g. `M1_Neuro_Programming`).
 - In VSCode, go to **File** → **Open Folder...** and select that folder.
 - Create a new file called `test.py`, type `print("Hello world")`, save it, then click the  **Run** button in the top-right corner of the editor.
 - You should see `Hello world` printed in the terminal panel at the bottom.

If this works, your setup is complete.

Part 2 — Creating a GitHub or GitLab account with your student email

2.1 Why do I need this?

GitHub and GitLab are online platforms for storing code, tracking changes over time, and collaborating with others (using a tool called **Git**). You will use one of them to submit and share your work during this master's program. You only need **one** of the two — ask your instructor which one the course will use if you're unsure.

Using your **university email address** (`prenom.nom@etu.univ-amu.fr` or similar) when you sign up gives you free access to extra benefits (private repositories, student developer packs, etc.), so it's worth using it instead of a personal address.

2.2 Creating a GitHub account

1. Go to github.com/signup.
2. Enter your **university email address**, choose a username, and set a password.

3. Verify your email: GitHub will send a confirmation code to your student inbox — check your inbox (and spam folder) and enter the code.
4. Skip or answer the optional setup questions (they don't matter for this course).
5. (Optional but recommended) Once your account is created, apply for the **GitHub Student Developer Pack** at education.github.com/pack using your student email — this unlocks free tools and services for students, though it is not required for the course itself.

2.3 Creating a GitLab account

1. Go to gitlab.com/users/sign_up.
2. Choose **"Register"** and fill in the form using your **university email address**.
3. Check your student inbox for a confirmation email and click the verification link.
4. Log in and complete your profile.

2.4 Good practices

- **Use your student email** so your account is recognizable as belonging to the university, and so you remain eligible for student benefits for as long as you're enrolled.
 - **Remember your username and password** (consider using a password manager) — you will need to log in regularly throughout the course.
 - Keep your account credentials private — never share your password with anyone.
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Next steps

Once you have:

- ☐ VSCodium installed and able to run a Python file
- ☐ Python and the VSCodium Python/Jupyter extensions installed
- ☐ A GitHub or GitLab account created with your student email

...you are ready for the "Introduction to Programming" session. If you run into any issue before the session, contact:

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